

Vision and Hearing

Macula (complement, ARMS2/HTRA1), glaucoma and intraocular pressure, myopia, crystalline, vitamin A of the retina, hearing and tinnitus

Sérgio F · 02-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. In the macula, Sérgio F does not have the risk alleles of ARMS2 A69S or HTRA1 (G/G in both), which form the second largest locus of DMI. The two DMI scores are typical (53.4 and 71.7 percentiles) and the CETP is favourable. In glaucoma, **TMCO1** (the strongest locus of intraocular pressure) is not at risk, and the intraocular pressure score is typical (63.3). The loci load of DMI is typical (33.2), and the hearing loss is also (56.1). It does not have the **MYOC Q368X** variant. The lens is typical (percentile 72.2 cataract, without risk alleles in the leading 2 loci). In hearing, there is no GJB2 M34T or V37I, and the loci KLHDC7B and CTBP2 are not at risk.

Unfavourable. The strongest point is the glaucoma: open angle consensus at the 85th percentile (127-variant score at 97.9; genomic score at 81.5). The glaucoma loci load is at percentile 93.7. The risk comes mainly from the vulnerability genes of the optic nerve (CDKN2B-AS1 A/A, SIX6 H141N C/C, TXNRD2 C/C), plus the GAS7 A/A. The intraocular pressure loci load is also high (percentile 94.6), although the pressure score is typical. Therefore, the risk may exist even with normal pressure. The second point is the myopia: the refractive error score is at percentile 7.6 (myopic trend), with GJD2, KCNQ5 and PRSS56 in risk homozygosity and myopia loci load at percentile 99.7. In the macula has CFH Y402H heterozygous, C3 R102G and CFI heterozygous and TIMP3 A/A (via the slightly unfavourable complement). In hearing, the score of use of hearing aids is at percentile 90.0, that of severity of tinnitus in 94.6 (loci load of tinnitus at 99.8), and ISG20/ACAN is C/C.

What does it mean? The eye to watch is the optic nerve: glaucoma is silent and visual field loss is irreversible. Therefore, it makes sense a complete ophthalmologic examination with OCT of the optic nerve, which detects glaucoma before symptoms. Myopia is added to this risk and also to retinal detachment. The macula has intermediate risk. It is the complement, without ARMS2, and lutein/zeaxanthin, omega-3 and zinc with copper cover it. Hearing has a moderate predisposition to loss with age and tinnitus, so protection against noise is the main lever.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (statistical imputation, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 37 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 31 of the chip, 6 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 4 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 11 PGS Catalog scores calculated locally with 876,845 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 2081 loci, 1871 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all scores, a higher percentile = more risk, except in the refractive error (high percentile = more hyperopic; low = more myopic).

Macula

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Macular degeneration of age	PGS004952 (Herold 2023, 52 variants)	46 52 (weighted coverage 70%)	53.4	TYPICAL	typical
Macular degeneration of age	PGS002269 (Zekavat 2022, 47 variants)	42 47 (weighted coverage 72%)	71.7	TYPICAL	typical

Glaucoma and intraocular pressure

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Glaucoma	PGS000137 (Craig 2020, 2673 variants)	2,385 2,673 (weighted coverage 84%)	81.5	MORE LIKELY	unfavourable
Primary open angle glaucoma	PGS002741 (Waksmunski 2022, 127 variants)	121 of 127 (weighted coverage 90%)	97.9 (consensus 85)	MORE LIKELY	unfavourable
Intraocular pressure	PGS000879 (Qassim 2020, 146 variants)	144 146 (weighted coverage 98%)	63.3	TYPICAL	typical

Crystalline

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Cataract	PGS001837 (Privé 2022, 11.6 thousand variants)	11,268 11,617 (weighted coverage 97%)	72.2	TYPICAL	typical

Refraction

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Refractive error (spherical equivalent)	PGS003860 (Clark 2023, Bayesian method, 768 thousand variants)	754,706 767,867 (weighted coverage 97%)	7.6	MORE MIOPICO	unfavourable

- **Refractive error (spherical equivalent):** High percentile = more hyperopic; low = more myopic. Myopia (especially high) increases the risk of glaucoma, retinal detachment and myopic maculopathy.

Hearing

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Hearing difficulties	PGS000762 (Cherny 2020, 100 thousand variants)	86,215 100,325 (weighted coverage 87%)	46.1	TYPICAL	typical
Use of hearing aids	PGS000763 (Cherny 2020, 4270 variants)	3,847 4,270 (weighted coverage 90%)	90.0	MORE LIKELY	unfavourable

Tinnitus

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Gravity of tinnitus	PGS001533 (Tanigawa 2022, 343 variants)	285 343 (weighted coverage 84%)	94.6	MORE LIKELY	unfavourable
Tinnitus	PGS019123 (Chen 2025, 18 thousand variants)	17,786 18,193 (weighted coverage 98%)	24.6	TYPICAL	typical (low reliability)

- **Tinnitus:** Developed in a cohort of another population: uncertain transferability, low reliability.

4. Key variants by topic

Variants with effect described in the literature. The judgment refers to the eye and the ear.

Macular degeneration of age (DMI)

Age macular degeneration (DMI) is dominated by two axes: the complement (CFH, C3, CFB, CFI) and the locus ARMS2/HTRA1. Next come the matrix of the Bruch membrane (TIMP3) and lipid metabolism (CETP, APOE).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CFH	rs1061170	Y402H	C/T	chip	C (1 copy/s)	unfavourable (light)	Y402H: the C allele (His402) reduces the binding of factor H to the retina surfaces, which gives more activation of the complement in the drusen. It is the main locus of DMI (OR ~2.4 per allele)	Strong
CFH	rs10737680		A/C	chip	A (1 copy/s)	unfavourable (light)	CFH leader variant in GWAS (OR ~2.4 by allele A)	Strong
ARMS2	rs10490924	A69S	G/G	chip	T (0 copy/s)	favourable	A69S: T allele is the second largest locus of DMI (OR ~2.8 per allele), especially in advanced/neovascular form	Strong
HTRA1	rs11200638		G/G	chip	A (0 copy/s)	favourable	HTRA1 promoter in imbalance with ARMS2: Neovascular DMI	Strong
C3	rs2230199	R102G	G/C	chip	C (1 copy/s)	unfavourable (light)	R102G: the allele 102G (C in strand +) gives more activation of the complement	Moderate
CFB	rs641153	R32Q	G/G	chip	A (0 copy/s)	neutral	R32Q: Allele A (Gln32) reduces the activation of the complement alternative pathway and protects from DMI (OR ~0.5)	Moderate
CFI	rs10033900		T/C	chip	T (1 copy/s)	unfavourable (light)	Complement factor I (OR ~1.2)	Moderate
TIMP3 (SYN3)	rs9621532		A/A	chip	A (2 copy/s)	unfavourable	Extracellular matrix of the Bruch membrane (OR ~1.4)	Moderate
CETP	rs3764261		C/C	chip	A (0 copy/s)	favourable	Metabolism of HDL: Allele A rises to HDL and is associated with more DMI (OR ~1.15)	Moderate
VEGFA (region)	rs943080		C/T	chip	T (1 copy/s)	unfavourable (light)	Angiogenesis (OR ~1.15)	Moderate
APOE	rs429358	ε4	T/T	imputation DR2=0.99	C (0 copy/s)	neutral	The ε4 (C allele) is protective for DMI, unlike the brain	Moderate
APOE	rs7412	ε2	C/C	chip	T (0 copy/s)	favourable	The ε2 (T allele) is associated with a little more DMI	Moderate

Glaucoma, optic nerve and intraocular pressure

Glaucoma is an optic neuropathy: retinal ganglion cells die, with or without high intraocular pressure. There are genes of pressure (TMCO1, CAV1, ABCA1, GAS7) and genes of the vulnerability of the optic nerve (CDKN2B-AS1, SIX6, TXNRD2). MYOC Q368X is a rare variant of large effect and LOXL1 marks the exfoliation syndrome.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CDKN2B-AS1 (9p21)	rs4977756		A/A	chip	A (2 copy/s)	unfavourable	Senescence of ganglion cells and optic nerve (OR ~1.4 by allele A)	Moderate
SIX6	rs33912345	H141N	C/C	chip	C (2 copy/s)	unfavourable	H141N: development of the optic nerve and retina (C allele increases the risk of glaucoma)	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SIX6 (region)	rs10483727		T/C	chip	T (1 copy/s)	unfavourable (light)	Larger optic cup and glaucoma (OR ~1.3)	Moderate
TMCO1	rs4656461		A/A	chip	G (0 copy/s)	favourable	Intraocular pressure ↑ and glaucoma (OR ~1.4–1.5)	Moderate
CAV1/CAV2	rs4236601		G/A	chip	A (1 copy/s)	unfavourable (light)	Caveolin: aqueous mood flow (OR ~1.3)	Moderate
ABCA1	rs2472493		G/A	chip	G (1 copy/s)	unfavourable (light)	Intraocular pressure ↑ and glaucoma	Moderate
GAS7	rs9913911		A/A	chip	A (2 copy/s)	unfavourable	Intraocular pressure ↑ and glaucoma	Moderate
AFAP1	rs4619890		A/A	chip	G (0 copy/s)	favourable	Open angle glaucoma (OR ~1.2)	Weak
TXNRD2	rs35934224		C/C	chip	C (2 copy/s)	unfavourable	Mitochondrial thioredoxin reductase: redox defense of ganglion cells (OR ~1.3)	Moderate
LOXL1	rs3825942	G153D	G/A	chip	G (1 copy/s)	typical	G153D: the G allele is at risk for exfoliation syndrome and exfoliative glaucoma (most of them have G/G)	Moderate
MYOC	rs74315329	Q368X	G/G	chip	A (0 copy/s)	favourable	Q368X: the most common pathogenic variant of the hereditary open angle glaucoma (rare)	Strong

Myopia and growth of the eye

Myopia comes mainly from the excessive axial growth of the eye. The loci involved act in retinal signaling (GJD2, KCNQ5), sclera (LAMA2) and eye growth (PRSS56).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
GJD2	rs524952		A/A	chip	A (2 copy/s)	unfavourable	Retinal communication junctions: allele A is associated with more myopia	Moderate
LAMA2	rs12193446		A/G	chip	A (1 copy/s)	unfavourable (light)	Sclera laminin: A allele is associated with more myopia	Moderate
KCNQ5	rs7744813		A/A	chip	A (2 copy/s)	unfavourable	Potassium channel of the pigment epithelium: allele A is associated with more myopia	Moderate
PRSS56	rs1550094		G/G	chip	G (2 copy/s)	unfavourable	Axial growth of the eye: G allele is associated with more myopia	Moderate

Crystalline (cataract)

Two loci of cataract (calcium and crystalline homeostasis).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SLC24A3	rs4814857		A/A	imputation DR2=0.97	G (0 copy/s)	favourable	Calcium transport in the lens: cataract (OR ~1.09)	Moderate
Intergenic 7p (ZNF619P1)	rs17172647		A/A	imputation DR2=0.96	G (0 copy/s)	favourable	Leader locus of cataract (OR ~1.07)	Weak

Vitamin A and retinal pigments

BCO1 converts beta-carotene into retina, the photoreceptor chromophore. The lutein and zeaxanthin form the macular pigment without going through BCO1.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
BCO1	rs12934922	R267S	A/T	chip	T (1 copy/s)	unfavourable (light)	R267S: lower conversion of beta-carotene to vitamin A (retinal)	Moderate
BCO1	rs7501331	A379V	C/C	chip	T (0 copy/s)	favourable	A379V: lower conversion of beta-carotene to vitamin A; the effect is added to R267S	Moderate

Hearing

GJB2 (connectin 26) is the most common genetic cause of recessive deafness. The others are GWAS loci of hearing loss of age: hair cells (KLHDC7B, ILDR1), tape synapse (CTBP2) and ISG20/ACAN region.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
GJB2 (connectin 26)	rs35887622	M34T	A/A	chip	G (0 copy/s)	favourable	M34T: mild recessive hearing loss variant (low penetration)	Moderate
GJB2 (connectin 26)	rs72474224	V37I	C/C	chip	T (0 copy/s)	favourable	V37I: mild to moderate recessive hearing loss variant	Moderate
ISG20/ACAN	rs12594617		C/C	imputation DR2=1.00	C (2 copy/s)	unfavourable	Head locus of sensorineural hearing loss	Moderate
KLHDC7B	rs36062310		G/G	imputation DR2=0.90	A (0 copy/s)	favourable	Hearing loss of age (OR ~1.15): outer hair cells	Moderate
CTBP2	rs10901863		C/C	imputation DR2=0.80	T (0 copy/s)	favourable	Hearing loss of age (synapsis in internal hair cell)	Moderate
ILDR1	rs2877561		A/C	chip	A (1 copy/s)	unfavourable (light)	Hearing loss of age (OR ~1.14)	Moderate

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of each trait, with each allele oriented towards risk (or more intraocular pressure), compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Macular degeneration of age	181 (167)	155 of 334	159	33.2	typical
Glaucoma	885 (840)	879 of 1680	836	93.7	unfavourable
Intraocular pressure	579 (511)	562 of 1022	509	94.6	unfavourable
Myopia	105 (96)	106 of 192	96	99.7	unfavourable
Cataract	134 (97)	75 of 194	97	77.7	typical
Hearing loss	143 (116)	101 of 232	116	56.1	typical
Tinnitus	54 (44)	58 of 88	44	99.8	unfavourable

6. Favourable

- **ARMS2 A69S G/G and HTRA1 G/G**: without the second largest locus of DMI (OR ~2.8 per allele). Therefore, the risk of advanced/neovascular DMI is close to the typical one, despite CFH. **Strong.**
- **Typical DMI scores**: percentiles 53.4 and 71.7 (52 and 47 variants; weighted coverage ~70%). **Moderate.**
- **CETP rs3764261 C/C**: without the allele that rises HDL and is associated with DMI. Moderate. APOE without $\epsilon 2$ ($\epsilon 3/\epsilon 3$), which is slightly at risk for DMI.
- **MYOC Q368X absent** (G/G): without the most common variant of glaucoma. Strong for the absence of this variant; the chip does not cover the other rare variants of MYOC.
- **TMC01 rs4656461 A/A**, the strongest locus of intraocular pressure and glaucoma, is without risk. Typical intraocular pressure score (percentile 63.3). AFAP1 also without risk. **Moderate.**
- **Crystalline**: cataract at percentile 72.2 (typical; loci load at 77.7); SLC24A3 and the locus 7p without risk alleles. **Moderate.**
- **GJB2 without M34T or V37I** (A/A and C/C). Moderate. KLHDC7B G/G and CTBP2 C/C without hearing loss alleles of age. Typical hearing impairment score (percentile 46.1). **Weak moderate.**
- **BCO1 A379V absent** (C/C). With the heterozygous R267S, beta-carotene conversion is only slightly reduced. **Weak.**

7. Unfavourable

- **Open angle glaucoma:** consensus at the 85th percentile (single scores: 97.9 for the 127-variant score, weighted coverage 90%; 81.5 for the 2673-variant score). The glaucoma loci load is at percentile 93.7. It is the most relevant finding of the report. **Moderate-strong.**
- **Optical nerve genes:** **CDKN2B-AS1 rs4977756 A/A** (9p21: senescence of ganglion cells; OR ~1.4 per allele), **SIX6 H141N C/C** (development and aging of the optic nerve), **TXNRD2 C/C** (mitochondrial redox defense of ganglion cells) and SIX6 rs10483727 T/C. They are vulnerability genes, not pressure genes. **Moderate.**
- **Intraocular pressure:** loci load at percentile 94.6, with GAS7 A/A and CAV1/CAV2 and ABCA1 heterozygotes. The pressure score (146 variants) is typical (63.3), so the signal is mixed. **Weak moderate.**
- **LOXL1 G153D G/A:** intermediate risk of exfoliation syndrome. Weak-moderate (G is very common).
- **Myopia:** refractive error at percentile 7.6 (768 thousand variants; myopic trend). Load of myopia loci at percentile 99.7. GJD2 A/A, KCNQ5 A/A, PRSS56 G/G and LAMA2 A/G. Myopia increases the risk of glaucoma, retinal detachment and myopic maculopathy. **Moderate.**
- **Complement pathway in macula:** **CFH Y402H C/T** (OR ~2.4 by allele; heterozygous) and CFH rs10737680 A/C. Also C3 R102G and CFI heterozygotes, TIMP3 A/A and VEGFA T/C. Moderate. The lack of ARMS2 attenuates the set.
- **Hearing:** use of hearing aids at percentile 90.0; ISG20/ACAN rs12594617 C/C (leading locus of sensorineural loss); ILDR1 A/C. **Moderate.**
- **Tinnitus:** severity at percentile 94.6 (343 variants) and loci load at percentile 99.8. The other score of tinnitus (percentile 24.6) was developed in another population and has low reliability. **Weak moderate.**
- **BCO1 R267S A/T:** conversion of slightly reduced beta-carotene. **Weak.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Cell Senescence	CDKN2B-AS1 (9p21) A/A: the locus p15/p16 associated with the senescence of retinal ganglion cells and glaucoma	Unfavourable
Redox imbalance	TXNRD2 C/C (thioredoxin reductase mitochondrial ganglion cells); BCO1 R267S heterozygous	Unfavourable (mild)
Mitochondrial dysfunction	TXNRD2 C/C: less mitochondrial antioxidant defense in the optic nerve	Unfavourable (mild)
Chronic Inflammation	Retinal complement: CFH Y402H C/T, C3 R102G G/C and CFI T/C (1 risk allele each); CFB without protective allele	Unfavourable (mild)
Inflamaging	The same route of complement in the drusen; ARMS2/HTRA1 without attenuation risk	Unfavourable (mild)
Protein Aggregation	Drusen (complement) and exfoliation material (LOXL1 G/A): intermediate risk	Neutral to unfavourable (mild)
Remodelling of the extracellular matrix	TIMP3 A/A (Broch's membrane); LAMA2 A/G (sclera: myopia); CAV1 and GAS7 (trabecular mesh)	Unfavourable (mild)
Fragmentation of Elastin	LOXL1 G153D G/A (Elastin cross-linking; exfoliation)	Neutral
Intercellular Communication	GJD2 A/A (retinal communication junctions: myopia); GJB2 without variants of deafness (cochlea junctions)	Mixed
Vascular Aging	CETP C/C and APOE ε3/ε3 (favourable for macula); VEGFA T/C	Neutral
Accumulation of Advanced Glycation Final Products (AGEs)	Crystalline: cataract at percentile 72.2 (typical; crystalline glycation is the main pathway)	Neutral
Exhaustion of Stem Cells	Cochlea ciliated cells do not regenerate: hearing loss of age (auditory device at percentile 90.0); ISG20/ACAN C/C	Unfavourable (mild)

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Full baseline ophthalmologic examination, then every 1-2 years:** Intraocular pressure, OCT of the optic nerve and layer of nerve fibers, visual field and bottom of the eye with dilation. The genetic risk of glaucoma (percentile ~98) includes the vulnerability of the optic nerve itself, so a normal pressure is not enough to exclude it. Suggestion (not applied).
2. **OCT of the macula in the same exam.** Test with the Amsler grid (each eye, 1x/month) from the 50 years or earlier if there are drusen. Suggestion (not applied).
3. **Myopia:** knowing the signs of retinal detachment (sudden fly flies, flashes, curtain in the visual field), which ask for observation on the day itself. In correction, prefer lenses that do not leave the myopia subcorrected. Suggestion (not applied).
4. **Lutein 20 mg + zeaxanthin 4 mg, omega-3 3.2-4 g/day and total zinc 20-25 mg with copper (already in the plan).** They cover the complement/CFH pathway of the macula; it is not necessary to change. Dark green leaves. Suggestion (not applied).
5. **Regular aerobic exercise:** low intraocular pressure and improves the optic nerve flow. Avoid prolonged positions from head down and Valsalva strong and repeated in heavy lifting (follow the intraocular pressure acutely). Suggestion (not applied).
6. **Caffeine $\leq$ 200 mg only in the morning (as in the plan):** High doses increase intraocular pressure. Suggestion (not applied).
7. **Eye protection:** sun glasses with an open air UV400 filter (crystalline and macula). Isotretinoin can dry the eye, so artificial tears if necessary. Suggestion (not applied).
8. **Protecting hearing** (auditory device in the ~the 90th percentile; ISG20/ACAN C/C: foam or music buffers in concerts, noisy gymnasiums and tools; 60/60 rule on the headphones ($\leq$ 60% of the volume, $\leq$ 60 min followed). Hearing loss is also one of the greatest modifiable factors of cognitive decline. Suggestion (not applied).
9. **Base audiogram with high frequencies,** then every 2-3 years. An early loss is corrected with an early device. Suggestion (not applied).
10. **Tinnitus** (gravity in the ~the 95th percentile; loci load ~99): to protect from noise, regular sleep and stress management. If they appear, evaluate early (audiogram); sound therapy and TCC are the most evident approaches. Suggestion (not applied).
11. **Otototoxic:** ask for alternatives to aminoglycosides and diuretics in high dose when there is an option. The m.1555A>G (MT-RNR1, sensitivity to aminoglycosides) is not on the chip. Suggestion (not applied).
12. **Retinal and cochlea vessels:** LDL/ApoB as low as possible (the lower, the better), high fiber (the better) and stable glycemia. Suggestion (not applied).
13. **Citicoline (S38 of the plan):** has preliminary evidence of support for ganglion cells in glaucoma, and is another reason to maintain it if adopted. Suggestion (not applied).

Not recommended: retinol or cod liver oil (isotretinoin; BCO1 does not justify it); beta-carotene in high dose; telomerase activators. Pausing curcumin, luteolin, apigenin and quercetin while the ALT is high is a suggestion (not applied; the real stack has not changed); the monthly senolytic is only fisetin + quercetin and only with normal ALT.

10. Limitations

- A genotyping chip with imputation does not read all variants. Most of the rare variants of MYOC, OPTN, TBK1 and CYP1B1 (glaucoma), the rare variants of CFH, CFI and C3 (large effect DMI), the retinal dystrophies (ABCA4, RPGR, USH2A...) and the GJB2 35delG deletion (not reliably evaluated in this chip). The m.1555A>G of the mitochondrial DNA is not in the data.
- Glaucoma and intraocular pressure scores explain a relevant part of the risk, but do not replace the examination of the optic nerve. The refractive error predicts trend, not the exact graduation.
- Hearing scores and tinnitus are based on self-declaration (auditory difficulty, use of hearing aids, tinnitus), so they have more noise than an audiogram.
- The score of tinnitus PGS019123 was developed in another population, and its calibration for Sérgio F is uncertain.
- The load of loci GWAS sums SNPs in linkage imbalance. It is a descriptive complement and does not replace the polygenic scores.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local statistical imputation (public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The alleles of each location (reference/alternative) are read from the Imputed VCF, which includes the SNPs of the chip. The published effect allele is compared with those alleles. If not coincided, the complement is used (only in non-palindromic SNPs). In palindromic SNPs (C3 rs2230199, GJD2 rs524952, BCO1 rs12934922, ISG20/ACAN rs12594617) the strand + (standard of GWAS Catalog and harmonized files of PGS Catalog) is assumed.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog: PGS004952 (Herold 2023) and PGS002269 (Zekavat 2022) for DMI; PGS000137 (Craig 2020) and PGS002741 (Waksmunski 2022) for glaucoma; PGS000879 (Qassim 2020) for intraocular pressure; PGS001837 (Privé 2022) for cataract; PGS003860 (Clark 2023) for the refractive error PGS000762 and PGS000763 (Cherny 2020) for hearing; PGS001533 (Tanigawa 2022) and PGS019123 (Chen 2025) for tinnitus. Score = Δ coefficient $\times$ dose of effect allele. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog (associations with $p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** GWAS Catalog, ClinVar, dbSNP/Ensembl and reference literature: Fritsche 2016 (DMI), Gharahkhani 2021 and Craig 2020 (glaucoma), Hysi 2020 (refractive error), Wells 2019 (auditory), Clifford 2024 (acuphenes). The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Macular degeneration of age (DMI)	CFH	rs1061170	C/T	chip	C	1	strand +	1 copy of risk allele
Macular degeneration of age (DMI)	CFH	rs10737680	A/C	chip	A	1	strand +	1 copy of risk allele
Macular degeneration of age (DMI)	ARMS2	rs10490924	G/G	chip	T	0	strand +	without the risk allele
Macular degeneration of age (DMI)	HTRA1	rs11200638	G/G	chip	A	0	strand +	without the risk allele
Macular degeneration of age (DMI)	C3	rs2230199	G/C	chip	C	1	strand + (palindromic; assumed +)	1 copy of risk allele
Macular degeneration of age (DMI)	CFB	rs641153	G/G	chip	A	0	strand +	without the protective allele (common genotype)
Macular degeneration of age (DMI)	CFI	rs10033900	T/C	chip	T	1	strand +	1 copy of risk allele
Macular degeneration of age (DMI)	TIMP3 (SYN3)	rs9621532	A/A	chip	A	2	strand +	2 risk allele copies
Macular degeneration of age (DMI)	CETP	rs3764261	C/C	chip	A	0	strand +	without the risk allele
Macular degeneration of age (DMI)	VEGFA (region)	rs943080	C/T	chip	T	1	strand +	1 copy of risk allele
Macular degeneration of age (DMI)	APOE	rs429358	T/T	imputation DR2=0.99	C	0	strand +	without ε4 (without macular protection of ε4)
Macular degeneration of age (DMI)	APOE	rs7412	C/C	chip	T	0	strand +	without the risk allele
Glaucoma, optic nerve and intraocular pressure	CDKN2B-AS1 (9p21)	rs4977756	A/A	chip	A	2	strand +	2 risk allele copies
Glaucoma, optic nerve and intraocular pressure	SIX6	rs33912345	C/C	chip	C	2	strand +	2 risk allele copies
Glaucoma, optic nerve and intraocular pressure	SIX6 (region)	rs10483727	T/C	chip	T	1	strand +	1 copy of risk allele
Glaucoma, optic nerve and intraocular pressure	TMC01	rs4656461	A/A	chip	G	0	strand +	without the risk allele
Glaucoma, optic nerve and intraocular pressure	CAV1/CAV2	rs4236601	G/A	chip	A	1	strand +	1 copy of risk allele

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Glaucoma, optic nerve and intraocular pressure	ABCA1	rs2472493	G/A	chip	G	1	strand +	1 copy of risk allele
Glaucoma, optic nerve and intraocular pressure	GAS7	rs9913911	A/A	chip	A	2	strand +	2 risk allele copies
Glaucoma, optic nerve and intraocular pressure	AFAP1	rs4619890	A/A	chip	G	0	strand +	without the risk allele
Glaucoma, optic nerve and intraocular pressure	TXNRD2	rs35934224	C/C	chip	C	2	strand +	2 risk allele copies
Glaucoma, optic nerve and intraocular pressure	LOXL1	rs3825942	G/A	chip	G	1	strand +	G/A: intermediate risk
Glaucoma, optic nerve and intraocular pressure	MYOC	rs74315329	G/G	chip	A	0	strand +	without the risk allele
Myopia and growth of the eye	GJD2	rs524952	A/A	chip	A	2	strand + (palindromic; assumed +)	2 risk allele copies
Myopia and growth of the eye	LAMA2	rs12193446	A/G	chip	A	1	strand +	1 copy of risk allele
Myopia and growth of the eye	KCNQ5	rs7744813	A/A	chip	A	2	strand +	2 risk allele copies
Myopia and growth of the eye	PRSS56	rs1550094	G/G	chip	G	2	strand +	2 risk allele copies
Crystalline (cataract)	SLC24A3	rs4814857	A/A	imputation DR2=0.97	G	0	strand +	without the risk allele
Crystalline (cataract)	Intergenic 7p (ZNF619P1)	rs17172647	A/A	imputation DR2=0.96	G	0	strand +	without the risk allele
Vitamin A and retinal pigments	BCO1	rs12934922	A/T	chip	T	1	strand + (palindromic; assumed +)	1 copy of risk allele
Vitamin A and retinal pigments	BCO1	rs7501331	C/C	chip	T	0	strand +	without the risk allele
Hearing	GJB2 (connectin 26)	rs35887622	A/A	chip	G	0	strand +	without the risk allele
Hearing	GJB2 (connectin 26)	rs72474224	C/C	chip	T	0	strand +	without the risk allele
Hearing	ISG20/ACAN	rs12594617	C/C	imputation DR2=1.00	C	2	strand + (palindromic; assumed +)	2 risk allele copies
Hearing	KLHDC7B	rs36062310	G/G	imputation DR2=0.90	A	0	strand +	without the risk allele
Hearing	CTBP2	rs10901863	C/C	imputation DR2=0.80	T	0	strand +	without the risk allele
Hearing	ILDR1	rs2877561	A/C	chip	A	1	strand +	1 copy of risk allele